

SEAS BP8

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1 Linear system

The system to solve for $\mathbf{u}$ reads

$$-HP(D + SAT)P\mathbf{u} = HP(D + SAT)\chi(\vec{s}) \quad (1)$$

- H is the SBP norm. Here it is only included to symmetrize the matrix so that CG works.
- P is a projection operator that averages slip in the tangential components ($j = 2, 3$) on the fault. That is, $P\mathbf{u}$ replaces both u_j^+ and u_j^- by the average $\frac{u_j^+ + u_j^-}{2}$ on the fault, for $j = 1, 2, 3$. P also sets the Dirichlet condition $u_j = 0, j = 1, 2, 3$ at farfield boundaries.
- D is the usual elastic operator, without SATs (or projections).
- SAT couples tractions on the fault. It is the usual SAT for traction-free BC, but with the traction from the other side as data. It can be applied on only one side (it does not matter which one because the subsequent multiplication by P will average it out). Alternatively, half can be applied on both sides.
- $\chi(\vec{s})$ is where the current slip $\vec{s}$ comes in. $\chi(\vec{s})$ is a vector of the same size as the solution vector $\mathbf{u}$. It is zero everywhere except on the fault. The u_1 components are also zero. For $j = 2, 3$, the $+$ side gets value $s_j/2$, and the $-$ side value $-s_j/2$ so that the difference is s_j .

Some observations:

- The "true" displacement vector is $\mathbf{U} = P\mathbf{u} + \chi(\vec{s})$. $\mathbf{U}$ has the correct jump across the fault. Later, in the friction law, tractions should be computed using $\mathbf{U}$, not $\mathbf{u}$.
- The system matrix $-HP(D + SAT)P$ should be positive semidefinite. (Normally CG requires definite, not semidefinite, but since the RHS is also projected this might be okay?)
- The system comes from

$$P(D + SAT)\mathbf{U} = 0 \quad (2)$$

after substituting and rearranging, and multiplying by H .